Coherent DOAs estimation via muti-task network enhanced sparse variational Bayesian method

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***Abstract*—Direction of arrival (DOA) estimation is a crucial component of array signal processing. Classical DOA estimation algorithms, such as the multiple signal classification (MUSIC) and the estimation of signal parameters via rotational invariance techniques (ESPRIT), typically perform angle estimation under the assumption that sources are mutually independent of each other, which is often violated in practice due to the effects of, e.g., multi-path, leading to the coherent nature of sources at the receiver. To address this problem, this paper proposes a joint scheme integrating deep learning networks and sparse variational Bayesian (SVB). Specifically, the proposed method first utilises a deep learning network to extract angle features from the received signals, yielding the correlation relationships between sources and coarse estimates of DOA. Subsequently, accurate DOA estimation is achieved via SVB.**

Simulation results demonstrate that the proposed method can effectively cope with the impacts of complex gain errors and signal multipath effects, and its performance outperforms that of the traditional MUSIC algorithm.

***Index Terms*—DOA estimation, coherent sources, deep learning, convolution neural network, sparse variational Bayesian.**

I. Introduction

Direction of Arrival (DOA) estimation is one of the research hotspots in the field of array signal processing, and it has been widely applied in numerous areas [1] such as target tracking [2], navigation and positioning [3], and underwater navigation [4]. Among traditional DOA estimation methods, subspace-based approaches (e.g., the Multiple Signal Classification algorithm [5] and the Estimation of Signal Parameters via Rotational Invariance Techniques algorithm [6]) are usually constructed based on complex mathematical models. Besides the aforementioned subspace-based algorithms, traditional DOA estimation methods also include conventional beamforming [7], the minimum variance distortionless response (MVDR) method [8], the classical maximum likelihood (ML) DOA estimation method [9], and the weighted subspace fitting (WSF) method [10], among others. They can achieve high-precision angle measurement under ideal conditions, yet they impose stringent requirements on the number of snapshots and computational complexity. In practical applications, interfering factors such as coherent signal sources and array amplitude-phase errors exist. It is noticed that the coherent signal sources include those generated by multipath effects.

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In recent years, numerous machine learning-based methods [11–14] have been proposed for the DOA estimation problem, which leverage neural networks to extract features and perform end-to-end training to estimate DOA. [15] proposed a convolutional neural network (CNN)-based method to address the performance limitations of traditional algorithms constrained by noisy and multipath propagation environments. [16] introduced deep neural network (DNN), 1-dimensional convolutional neural network (1-D CNN) and 2-dimensional convolutional neural network (2-D CNN) models, along with their optimisation strategies, to mitigate the phase distortion caused by multipath signals. Although the aforementioned data-driven methods can address the problem of poor performance of traditional subspace-based methods under certain non-ideal conditions, these data-driven methods suffer from low interpretability, high demand for large-scale labelled data, and high computational complexity, while generalisation capability also remains a critical issue.

To address the aforementioned issues, subsequent studies have proposed hybrid methods combining data-driven and model-driven approaches for DOA estimation. For instance, the SubspaceNet method [17] [18] trains a subspace network to transform the received signal covariance matrix under non-ideal conditions into an equivalent covariance matrix, which can then be fed into the Root-MUSIC algorithm [19]to achieve DOA estimation in non-ideal scenarios. Another approach is the differentiable MUSIC (diffMUSIC) algorithm [20], replaces the non-differentiable argmax function in the spectral peak search stage of the traditional MUSIC algorithm with a differentiable softmax function, thereby solving the problems of antenna array position errors and complex gain errors. In [21], the expectation-maximisation (EM) algorithm was employed to update the probability distribution parameters, and the two types of error parameters were solved alternately to achieve accurate DOA estimation results.

In this paper, we are going to address the problem of DOA estimation for coherent sources. The proposed method is made up of two phases. First of all, a multi-task network is designed to extract the features of received signals, including coherence relationships, coarse angle estimation, and signal power. Then the sparse variational Bayesian (SVB) is leveraged to perform DOA estimation, where the coarse angle estimation output by the network is taken as the reference to refine the angular grid for accurate angle estimation. Compared to the EM algorithm that maximises the likelihood function, VBI provides the posterior of source DOAs, thus having a larger performance upper limit. A remarkable feature of the proposed algorithm is that it can overcome the effect caused by the complex gain error. The performance of the proposed algorithm is verified via simulations.

II. Problem formulation

Under ideal conditions, the received signal model of a uniform linear array (ULA) consisting of K array elements is generally regarded as

$$\mathbf{X} = \mathbf{A}(\theta)\mathbf{S} + \mathbf{N}, \tag{1}$$

$\mathbf{X} \in \mathbb{C}^{K\times T}$denote the received signal,where T is the number of snapshots.The array manifold $\mathbf{A}(\theta)$ is expressed as

$$\mathbf{A}(\theta) = \left[\mathbf{a}(\theta_1), \mathbf{a}(\theta_2), \dots, \mathbf{a}(\theta_P)\right], \tag{2}$$

where P is the number of sources, and

$$\begin{aligned}\mathbf{a}(\theta_i) = \big[1 \; & e^{-j\frac{2\pi d}{\lambda}\sin\theta} \; e^{-j\frac{2\pi d}{\lambda}2\sin\theta} \; \dots \\ & e^{-j\frac{2\pi d}{\lambda}(K-1)\sin\theta}\big]^{\mathrm{T}},\end{aligned} \tag{3}$$

where $\mathbf{N}$ denotes the white Gaussian noise, and its distribution satisfies

$$\mathbf{N} \sim \mathcal{CN}(\mathbf{0}, \sigma_n^2\mathbf{I}). \tag{4}$$

The aforementioned content describes the signal model of a uniform linear array in the ideal case. Nevertheless, the actual received signal of the array is inevitably affected by non-ideal factors, which contain array element position errors, antenna complex gain mismatches and coherent signal sources. At this time, the actual signal model can be expressed as

$$\mathbf{X} = \mathbf{\Gamma}\mathbf{A}(\theta)\mathbf{S}' + \mathbf{N}, \tag{5}$$

where $\mathbf{\Gamma} \in \mathbb{C}^{K\times K}$ denotes the complex gain error matrix of the antenna array. $\boldsymbol{S'}$ denotes coherent signal sources. The complex gain error matrix $\mathbf{\Gamma}$ is a diagonal matrix, which is expanded as

$$\mathbf{\Gamma} = \begin{bmatrix} \gamma_1 & 0 & \dots & 0 \\ 0 & \gamma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \gamma_K \end{bmatrix} \in \mathbb{C}^{K\times K}, \tag{6}$$

where γ_k denotes the complex gain (or fading coefficient) of the k-th array element. Following the physical characteristics of hardware impairments, γ_k is modelled as a small perturbation around the ideal value 1 (i.e., the Rician-type complex representation with line-of-sight (LOS) component), and its probability model is given by

$$\gamma_k \sim \mathcal{CN}\left(1, \frac{1}{\tau_k}\right), \tag{7}$$

where $\tau_k > 0$ is the precision (reciprocal of variance) of the k-th array element's complex gain, and τ_k quantifies the perturbation intensity of γ_k around 1. To enable the algorithm to adaptively learn the unknown fading intensity, we impose a Gamma hyperprior on τ_k

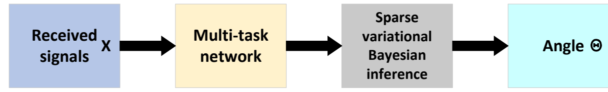

Fig. 1. Sketch of Proposed Flow Chart

$$\tau_k \sim \mathrm{Gamma}\left(\alpha_\tau, \beta_\tau\right), \tag{8}$$

with the expectation given by $\mathbb{E}[\tau_k] = \frac{\alpha_\tau}{\beta_\tau}$. When the received data indicates a large deviation of γ_k from 1, the posterior distribution of τ_k tends to be smaller (allowing stronger fluctuations of γ_k); conversely, τ_k becomes larger (pulling γ_k back to 1 more strongly). where $\mathbf{S}$ denotes the coherent signal sources.

III. The proposed DOA estimating algorithm for coherent sources

A. Sketch of proposed algorithm

The method proposed in this paper combines data-driven and model-driven approaches, as depicted in Fig. 1. As can be seen, the proposed algorithm is made up of two steps. First, the received signals are fed into a multi-task network to obtain the coherent topological relationship of signals, coarse angle estimation results, and power coefficients. The sparse variational Bayesian algorithm is employed for fine grid refinement on the coarse angle estimates to obtain accurate values. The output vector of the network topology branch is then multiplied by the angle vector from the traditional algorithm to screen true signal sources and generate fine-angle estimates, thereby achieving coherent signal resolution and complex gain error compensation simultaneously.

B. Multi-task learning network model

The multi-task learning network proposed in this paper takes the covariance matrix of the received data as input, aiming to simultaneously accomplish three tasks: topological mask prediction, coarse angle estimation, and power sensing, so as to achieve high-precision parameter estimation in coherent source scenarios.

As the core of feature extraction, the backbone network adopts a multi-layer convolutional neural network (CNN) structure. Based on the shared feature vectors output by the backbone network, three task-specific branch networks are designed to fulfill their respective learning objectives, where each branch employs a dedicated network structure and activation function adapted to the characteristics of the corresponding task.

The proposed multi-task network operates on the covariance matrix and comprises three dedicated branches, each built on two fully connected layers.

The topological mask branch uses a Sigmoid activation $\sigma(x)$ to constrain its output to $(0, 1)$, producing a spatial mask that predicts source existence probabilities and provides prior information for angle estimation.

The coarse angle estimation branch employs a Tanh activation $\tanh(x)$, mapping outputs to $(-1, 1)$ to directly regress source directions from phase-difference features, thereby offering initial estimates for refinement while lowering computational cost.

The power sensing branch adopts a ReLU activation $\text{ReLU}(x) = \max(0, x)$ to ensure non-negative power estimates, learning amplitude features to deliver prior power information for coherent source modelling.

We formally define the function of the backbone network as a composition of five successive layers

$$f(x) = f_5\big(f_4(f_3(f_2(f_1(x))))\big). \tag{9}$$

Building upon the shared features extracted by the backbone, the functions for the three task-specific branches are respectively defined as follows. The coherent topological mask prediction branch is represented by $g(x) = g_3(g_2(g_1(f(x))))$, the angle estimation branch by $h(x) = h_3(h_2(h_1(f(x))))$, and the power sensing branch by $k(x) = k_3(k_2(k_1(f(x))))$.

This functional formulation clearly illustrates a multi-task learning architecture where a common backbone $f(x)$ extracts shared representations, which are subsequently refined by dedicated branch networks $g(x)$, $h(x)$, and $k(x)$ to perform distinct sub-tasks.

C. Train and Inference

In this paper, the total loss function of the training part of the network is defined as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{mask}} + \lambda_1 \mathcal{L}_{\text{angle}} + \lambda_2 \mathcal{L}_{\text{power}}, \tag{10}$$

where $\mathcal{L}_{\text{mask}}$, $\mathcal{L}_{\text{angle}}$ and $\mathcal{L}_{\text{power}}$ are the loss functions of the topological mask prediction branch, coarse angle estimation branch and power sensing branch, respectively. λ_1 and λ_2 are the corresponding loss weight parameters. The loss function of the topological mask prediction branch adopts the binary cross-entropy (BCE) loss, which is defined as

$$\mathcal{L}_{\text{mask}} = \frac{1}{P^2} \sum_{i=1}^{P} \sum_{j=1}^{P} [y_{i,j}^{\text{mask}} \log(\hat{y}_{i,j}^{\text{mask}}) + (1 - y_{i,j}) \log(1 - \hat{y}_{i,j}^{\text{mask}})], \tag{11}$$

where y_{ij}^{mask} denotes the element of the coherence matrix between two signal sources. For example, $y_{12} = 1$ indicates that signal source 1 is coherent with signal source 2, and $y_{11} = 1$ means that a signal source is coherent with itself.

The loss function of the coarse angle estimation branch adopts the mean square error (MSE) loss, which is defined as

$$\mathcal{L}_{\text{angle}} = \frac{1}{P} \sum_{i=1}^{P} (\theta_i - \hat{\theta}_i)^2, \tag{12}$$

where $\hat{\theta}_i$ represents the estimated angle of the i-th signal source, and this MSE loss function is employed to characterise the squared difference between the estimated angle and the true angle.

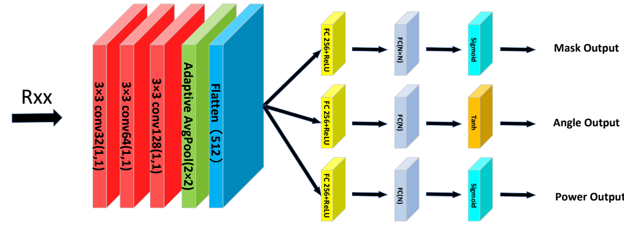

Fig. 2. Multi-task Network Flow Chart

The loss function of the power sensing branch also adopts MSE loss, which is defined as

$$\mathcal{L}_{\text{power}} = \frac{1}{P} \sum_{i=1}^{P} (p_i - \hat{p}_i)^2, \tag{13}$$

where $\hat{p}_i$ denotes the estimated power of the i-th signal source, and the MSE loss function is selected to learn the error between the true power and the estimated power of the signal source.

D. Sparse variational Bayesian inference

In this section, we integrate the previous results with the sparse Bayesian learning algorithm. We first introduce the dictionary matrix $\overline{\boldsymbol{A}} \in \mathbb{C}^{K \times M}$, which maps the M potential source directions to the antenna array.

The relationship between these quantities is given by the observation equation:

$$\boldsymbol{X} = \boldsymbol{\Gamma} \overline{\boldsymbol{A}} \boldsymbol{S'} + \boldsymbol{N} \quad \Longleftrightarrow \quad \boldsymbol{x}_t = \boldsymbol{\Gamma} \overline{\boldsymbol{A}} \boldsymbol{s}_t + \boldsymbol{n}_t. \tag{14}$$

The noise is modelled as complex Gaussian with a Gamma-distributed precision β:

$$\boldsymbol{n}_t \mid \beta \sim \mathcal{CN}\left(0, \beta^{-1} \boldsymbol{I}_K\right), \quad \beta \sim \text{Gamma}(c_0, d_0). \tag{15}$$

For the sparse matrix $\boldsymbol{S'}$, we adopt a row-sparse prior typical of SVB. Let the k-th row be denoted $\boldsymbol{s}_k \in \mathbb{C}^{1 \times T}$. Assuming independence across rows, we have:

$$\boldsymbol{s}_k \mid \alpha_k \sim \mathcal{CN}\left(0, \alpha_k^{-1} \boldsymbol{I}_T\right), \quad \alpha_k \sim \text{Gamma}(a_0, b_0). \tag{16}$$

This prior encourages row-wise sparsity: a larger α_k drives the entire k-th row toward zero.

The antenna gain errors γ_k are assumed to have a unit mean but unknown variance. We employ a standard hierarchical model where the precision η_k of each γ_k follows a Gamma prior:

$$\gamma_k \mid \eta_k \sim \mathcal{CN}\left(1, \eta_k^{-1}\right), \quad \eta_k \sim \text{Gamma}(e_0, f_0), \tag{17}$$

with independence across antennas.

The complete joint distribution of the model factors is:

$$\begin{aligned} p(\boldsymbol{X}, \boldsymbol{S'}, \boldsymbol{\alpha}, \boldsymbol{\gamma}, \boldsymbol{\eta}, \beta) = p(\boldsymbol{X} \mid \boldsymbol{S'}, \boldsymbol{\gamma}, \beta)\, p(\boldsymbol{S'} \mid \boldsymbol{\alpha})\, p(\boldsymbol{\alpha}) \\ \cdot\, p(\boldsymbol{\gamma} \mid \boldsymbol{\eta})\, p(\boldsymbol{\eta})\, p(\beta). \end{aligned} \tag{18}$$

The likelihood further factorises over snapshots:

$$p(\boldsymbol{X} \mid \boldsymbol{S'}, \boldsymbol{\gamma}, \beta) = \prod_{t=1}^{T} \mathcal{CN}\left(\boldsymbol{x}_t \mid \boldsymbol{\Gamma} \overline{\boldsymbol{A}} \boldsymbol{s}_t, \beta^{-1} \boldsymbol{I}_K\right). \tag{19}$$

To perform SVB, we assume a mean-field factorisation of the posterior:

$$q(\boldsymbol{S'}, \boldsymbol{\alpha}, \boldsymbol{\gamma}, \boldsymbol{\eta}, \beta) = q(\boldsymbol{S'})\, q(\boldsymbol{\alpha})\, q(\boldsymbol{\gamma})\, q(\boldsymbol{\eta})\, q(\beta). \tag{20}$$

The variational update follows the coordinate ascent rule. In particular, the updates for the noise precision β, the gain precision η, the gain γ, the sparsity parameter α, and the sparse coefficients $\boldsymbol{S'}$ are respectively given by

$$\ln q^{(l)}(\beta) = \int p(\boldsymbol{X}, \boldsymbol{S'}, \boldsymbol{\alpha}, \boldsymbol{\gamma}, \boldsymbol{\eta}, \beta) q^{(l-1)}(\boldsymbol{S'}) q^{(l-1)}(\boldsymbol{\alpha}) \times q^{(l-1)}(\boldsymbol{\gamma}) q^{(l-1)}(\boldsymbol{\eta})\, d\boldsymbol{S'}\, d\boldsymbol{\alpha}\, d\boldsymbol{\gamma}\, d\boldsymbol{\eta} + \text{const}, \tag{21}$$

$$\ln q^{(l)}(\eta) = \int p(\boldsymbol{X}, \boldsymbol{S'}, \boldsymbol{\alpha}, \boldsymbol{\gamma}, \boldsymbol{\eta}, \beta) q^{(l-1)}(\boldsymbol{S'}) q^{(l-1)}(\boldsymbol{\alpha}) \times q^{(l-1)}(\boldsymbol{\gamma}) q^{(l)}(\beta)\, d\boldsymbol{S'}\, d\boldsymbol{\alpha}\, d\boldsymbol{\gamma}\, d\beta + \text{const}, \tag{22}$$

$$\ln q^{(l)}(\gamma) = \int p(\boldsymbol{X}, \boldsymbol{S'}, \boldsymbol{\alpha}, \boldsymbol{\gamma}, \boldsymbol{\eta}, \beta) q^{(l-1)}(\boldsymbol{S'}) q^{(l-1)}(\boldsymbol{\alpha}) \times q^{(l)}(\eta) q^{(l)}(\beta)\, d\boldsymbol{S'}\, d\boldsymbol{\alpha}\, d\beta\, d\eta + \text{const}, \tag{23}$$

$$\ln q^{(l)}(\alpha) = \int p(\boldsymbol{X}, \boldsymbol{S'}, \boldsymbol{\alpha}, \boldsymbol{\gamma}, \boldsymbol{\eta}, \beta) q^{(l-1)}(\boldsymbol{S'}) q^{(l)}(\gamma) \times q^{(l)}(\eta) q^{(l)}(\beta)\, d\boldsymbol{S'}\, d\gamma\, d\eta\, d\beta + \text{const}, \tag{24}$$

$$\ln q^{(l)}(\boldsymbol{S'}) = \int p(\boldsymbol{X}, \boldsymbol{S'}, \boldsymbol{\alpha}, \boldsymbol{\gamma}, \boldsymbol{\eta}, \beta) q^{(l)}(\alpha) q^{(l)}(\gamma) \times q^{(l)}(\eta) q^{(l)}(\beta)\, d\alpha\, d\gamma\, d\eta\, d\beta + \text{const}, \tag{25}$$

where each expectation is taken over the respective set of other variables. The subsequent sections detail the derivation of the explicit form for each of these updates.

Algorithm 1 sparse variational Bayesian (SVB)

Input: Observation matrix $\mathbf{X}$, dictionary $\overline{\boldsymbol{A}}$, hyperparameters $\{a_0, b_0, c_0, d_0, e_0, f_0\}$, convergence threshold ε.
Output: Estimated DOA spectrum P_k and calibrated gains $\hat{\mathbf{\Gamma}}$.

repeat
  Update noise precision β: Compute the variational posterior of the noise precision parameter via (21).
  Update gain precision η: Obtain the variational update for the gain precision parameter using (22).
  Update gain γ: Derive the variational distribution of the gain parameter according to (23).
  Update sparsity parameter α: Calculate the variational posterior of the sparsity parameter following (24).
  Update sparse coefficients $\boldsymbol{S'}$: Estimate the variational posterior of the sparse coefficient matrix using (25).
until $\|\boldsymbol{\alpha}^{(l)} - \boldsymbol{\alpha}^{(l-1)}\| / \|\boldsymbol{\alpha}^{(l-1)}\| < \varepsilon$
Compute DOA spectrum: $P_k = \|\mathbf{s}_k\|^2 / T,\ k = 1, \ldots, M.$
Compute calibrated gains: $\hat{\mathbf{\Gamma}} = \mathrm{diag}(\bar{\gamma}_1, \ldots, \bar{\gamma}_K)$.
return P_k, $\hat{\mathbf{\Gamma}}$.

IV. Simulation Experiments

In the experiments, a ULA with $K = 20$ antennas is adopted, and the observation data is generated according to Eq. (5). The Direction of Arrival (DOA) of the sources is uniformly distributed within the range of $-60°$ to $60°$, and the variance of the antenna gain error is set to 0.01. Unless otherwise specified, the default experimental parameters are configured as follows: the number of sources $P = 4$ and the number of snapshots $T = 200$.

Fig. 3 illustrates the spatial spectrum of the proposed method at a signal-to-noise ratio (SNR) of -5 dB. Fig. 4 presents the comparison of root mean square error (RMSE) between the method proposed in this paper and the baseline MUSIC method under different SNRs, with the number of sources $P = 4$ and the number of snapshots $T = 200$. As can be seen from the simulation results, under the conditions of coherent sources and complex gain error, the proposed method exhibits superior DOA estimation performance compared to the baseline method across various SNR levels.

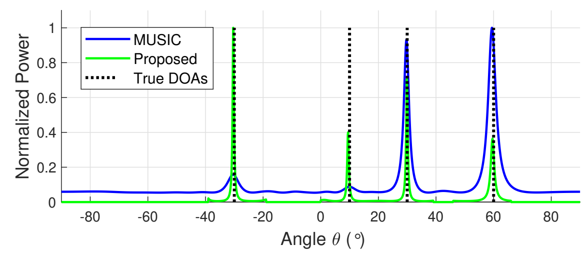

Fig. 3. DOA Estimation Comparison

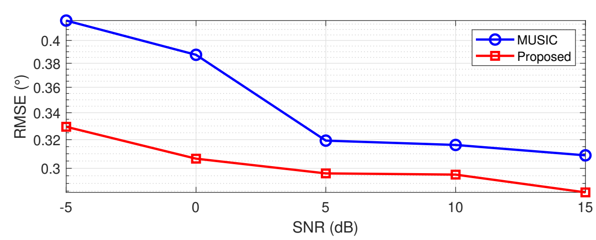

Fig. 4. RMSE under different signal-to-noise ratios (SNRs) with $T = 200$ snapshots.

V. Conclusion

This paper conducts research on the problems of coherent signal sources and complex gain error existing in received signals. A novel sparse variational Bayesian Direction of arrival (DOA) estimation method based on a multi-task network is proposed, which achieves decoherence and error correction simultaneously. Simulation results, in comparison with those of traditional methods, verify the feasibility of the proposed method. Future research will focus on introducing other non-ideal factors, such as position errors, into the proposed method.

References

[1] F. Jacob, J. Schmalenstroeer, and R. Haeb-Umbach, "Doa-based microphone array postion self-calibration using circular statistics," in *2013 IEEE International Conference on Acoustics, Speech and Signal Processing*. IEEE, 2013, pp. 116–120.

[2] J. Zhang, M. Bao, J. Yang, Z. Chen, and H. Hou, "Doa tracking algorithm based on avs pseudo-smoothing for coherent acoustic targets," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 59, no. 6, pp. 8175–8193, 2023.

[3] M. Pan, P. Liu, S. Liu, W. Qi, Y. Huang, X. You, X. Jia, and X. Li, "Efficient joint DOA and toa estimation for indoor positioning with 5g picocell base stations," *IEEE Transactions on Instrumentation and Measurement*, vol. 71, pp. 1–19, 2022.

[4] A. Liu, S. Shi, and X. Wang, "Robust doa estimation method for underwater acoustic vector sensor array in presence of ambient noise," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 61, pp. 1–14, 2023.

[5] R. Schmidt, "Multiple emitter location and signal parameter estimation," *IEEE transactions on antennas and propagation*, vol. 34, no. 3, pp. 276–280, 1986.

[6] R. Roy and T. Kailath, "Esprit-estimation of signal parameters via rotational invariance techniques," *IEEE Transactions on acoustics, speech, and signal processing*, vol. 37, no. 7, pp. 984–995, 2002.

[7] V. Krishnaveni, T. Kesavamurthy *et al.*, "Beamforming for direction-of-arrival (doa) estimation-a survey," *International Journal of Computer Applications*, vol. 61, no. 11, 2013.

[8] A. Aminu, M. Secmen, and M. A. Husein, "The effectiveness of jade based minimum variance distortionless response algorithm for doa estimation and adaptive beamforming in multipath smart antenna application," in *2014 IEEE 6th International Conference on Adaptive Science & Technology (ICAST)*. IEEE, 2014, pp. 1–7.

[9] M. Lavielle, E. Moulines, and J. Cardoso, "A maximum likelihood solution to doa estimation for discrete sources," in *Proc. Seventh IEEE Workshop on SP*, 1994, pp. 349–353.

[10] N. Hu, Z. Ye, D. Xu, and S. Cao, "A sparse recovery algorithm for doa estimation using weighted subspace fitting," *Signal processing*, vol. 92, no. 10, pp. 2566–2570, 2012.

[11] M. Chen, Y. Gong, and X. Mao, "Deep neural network for estimation of direction of arrival with antenna array," *IEEE Access*, vol. 8, pp. 140 688–140 698, 2020.

[12] J. Cong, X. Wang, M. Huang, and L. Wan, "Robust doa estimation method for mimo radar via deep neural networks," *IEEE Sensors Journal*, vol. 21, no. 6, pp. 7498–7507, 2020.

[13] W. Zhu and M. Zhang, "A deep learning architecture for broadband doa estimation," in *2019 IEEE 19th International Conference on Communication Technology (ICCT)*. IEEE, 2019, pp. 244–247.

[14] S. Chakrabarty and E. A. Habets, "Broadband doa estimation using convolutional neural networks trained with noise signals," in *2017 IEEE workshop on applications of signal processing to audio and acoustics (WASPAA)*. IEEE, 2017, pp. 136–140.

[15] H. Hassan, A. R. Maud, and M. Amin, "Deep learning based doa estimation in low snr and multipath scenarios," in *2024 IEEE International Symposium on Phased Array Systems and Technology (ARRAY)*, 2024, pp. 1–6.

[16] H. Xiang, B. Chen, T. Yang, and D. Liu, "Improved demultipath neural network models with self-paced feature-to-feature learning for doa estimation in multipath environment," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 5, pp. 5068–5078, 2020.

[17] D. H. Shmuel, J. P. Merkofer, G. Revach, R. J. Van Sloun, and N. Shlezinger, "Subspacenet: Deep learning-aided subspace methods for doa estimation," *IEEE Transactions on Vehicular Technology*, vol. 74, no. 3, pp. 4962–4976, 2024.

[18] A. M. Elbir, "Deepmusic: Multiple signal classification via deep learning," *IEEE Sensors Letters*, vol. 4, no. 4, pp. 1–4, 2020.

[19] B. D. Rao and K. S. Hari, "Performance analysis of root-music," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 37, no. 12, pp. 1939–1949, 2002.

[20] B. Chatelier, J. M. Mateos-Ramos, V. Corlay, C. Häger, M. Crussière, H. Wymeersch, and L. Le Magoarou, "Physically parameterized differentiable music for doa estimation with uncalibrated arrays," in *ICC 2025-IEEE International Conference on Communications*. IEEE, 2025, pp. 3858–3863.

[21] Y. Dong, Y. Xu, S. Liu, and M. Jin, "Doa estimation based on sparse bayesian learning under amplitude-phase error and position error," *Journal of Systems Engineering and Electronics*, vol. 36, no. 5, pp. 1122–1131, 2025.